

EXPERIENCE

Microsoft · Windows Fundamentals

Senior Software Engineer · Lead, Cross-Vertical AI Evaluation

2021–Present

AI EVALUATION FRAMEWORK

A production system where a frontier model analyzes Windows performance, power, and memory telemetry and proposes code optimizations engineers ship. I own the evaluation layer that decides which model suggestions are trustworthy.

- ▶ **Lifted production fix-acceptance from 54% to 78% on the frontier-model code-optimization pipeline by building its evaluation layer.** Currently the highest fix-acceptance rate among AI-assisted engineering workstreams in Windows. The framework evaluates whether model outputs are correct, grounded, actionable, and improving over time across Perf, Power, Memory, and future verticals on shared replay infrastructure.
- ▶ **Cut eval false positives ~35% and false negatives ~50% against a human-expert gold standard with a calibrated LLM-as-judge layer.** Scorers detect schema drift, parse failures, source-grounding errors, hallucinated APIs, unsupported mechanism and magnitude claims, classification inconsistencies, and overconfident recommendations. Judge calibration validated via Cohen's kappa.
- ▶ **Built reproducible replay infrastructure for model, prompt, advice-corpus, and context changes.** Each run holds inputs constant while varying one axis, then compares outputs across deterministic checks, cross-model disagreement, LLM-as-judge, human expert review, and production outcome tracking. Statistical methodology uses paired tests, bootstrap CIs, McNemar, Cohen's kappa, and Simpson's-paradox-aware aggregation, supporting model migration across Claude Opus (Anthropic SDK) and Azure-hosted GPT/o-series.
- ▶ **Defined the bi-directional taxonomy and feedback loop for AI-generated Windows optimizations.** Authored a taxonomy across algorithmic, caching, concurrency, I/O, memory, and anti-pattern classes; models use it for grounding, and novel model-proposed patterns route through validation gates before being admitted back into the corpus, creating a growing library of optimizations backed by shipped-fix evidence. Public post: [Bi-Directional Class-of-Fix Taxonomies](#). Architected the end-to-end loop (telemetry-driven bug discovery → prompt assembly → model analysis → evaluation → code-change generation → automated PR → regression testing) that was adopted as the team's technical direction.

WINDOWS GLITCHINESS METRIC

- ▶ **Designed and shipped Glitchiness, a new metric for perceived Windows responsiveness, validated with human-subjects studies.** Delay-signal end-time clustering with span-based duration and long-delay promotion, deployed across the Windows fleet with daily failure-rate CoV 4.8%. A 41-participant controlled perceptual study plus an n=2,234 in-the-wild user-feedback study show users report sluggishness +9.3pp more often when Glitchiness flags a bad period ($p=4.8e-5$, Cohen's $h=0.19$). The methodology of operationalizing an inherently fuzzy human-judgment concept into a statistically-validated measurement transfers directly to evaluating subjective, hard-to-specify qualities of AI systems.
- ▶ **Connected memory pressure to perceived OS responsiveness at fleet scale.** Built an algorithm joining the Glitchiness device-day base to system memory telemetry, quantifying how memory pressure degrades perceived responsiveness with a clear dose-response in daily usage and reproducing the effect across CPU families and device populations. The finding is informing Windows memory-management strategy.

FLEET TELEMETRY & FORECASTING SYSTEMS

- ▶ **Led metric and forecasting systems for Windows performance decisions and rebuilt the underlying fleet telemetry infrastructure.** Created the LongInputDelays leadership metric with a normalized hits-per-million-devices framing that avoids population-shift artifacts. Built the SLT Glidepath forecasting system (least-squares regression on log-transformed census, two-phase hybrid growth model, $R^2=0.90$ backtested across active silicon). Rewrote PerfTrack targeting/routing after finding a measurement blind spot, growing retail performance telemetry from hundreds of thousands to millions of devices/day and multiplying the signal underneath every downstream Windows performance metric.
- ▶ **Earlier work (2021–2024).** Foundational telemetry-driven regression-detection platform that became the substrate for the current evaluation work; cross-org device-targeting pipeline scaled from 100K to 2.5M devices; executive metric system with distance-to-target attribution and predictive forecasting; population-shift detection separating product regressions from fleet-composition changes; introduction of mouse/keyboard latency as a first-class telemetry signal that Glitchiness was later built on.

Apple · Google

AI/ML & Software Engineering Intern

Summers 2017–2019

- ▶ **Apple Siri Search (2019):** expanded Knowledge Card entity-type coverage and built a Hive/Impala automation pipeline for query-log-driven inline card refresh. **+214% entity coverage, +35% positive abandonment, +7% engagement** via A/B tests. **Google Cloud (2017):** multithreaded gRPC load generator and scheduler for next-generation network systems, **+31% network performance**. **Google Search (2018):** backend infrastructure for a food-ordering feedback system (Spanner schema migration, encrypted ingestion APIs, real-time partner-rating dashboards).

SKILLS

ML / AI Evaluation: LLM-as-judge calibration, scorer design, replay infrastructure, sycophancy / hallucination / scorer-gaming detection, frontier-model comparison (Claude Opus via Anthropic SDK, GPT-5, o-series), human-subjects studies, perceptual psychophysics.

Statistics: paired hypothesis testing, bootstrap CIs, McNemar, Cohen's kappa & h, Brier score, Simpson's-paradox-aware aggregation, calibration vs. human gold standard, time-series forecasting.

Systems: Python, C# / .NET 10 (Blazor Server, EF Core), TypeScript, SQL, Kusto/KQL, USQL/Scope (Azure Data Lake Analytics), Azure OpenAI, GitHub Copilot SDK, MCP, Azure Batch, Azure SQL.